WORKED SOLUTION

STEP II 1999, Question 5

Pure • Trigonometric Equations

STEP II 1999, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q5

The key idea

There are two natural routes. One is to make $\cos\alpha$ the unknown: isolate $\sin\alpha$, square, and use $\sin^2 + \cos^2 = 1$ to get a quadratic in $\cos\alpha$. The other is to write $5\cos x + 12\sin x$ as a single cosine, $13\cos(x-\beta)$, which also shows that both roots of the quadratic really occur. For the final part, $\alpha < \frac{3}{4}\pi$ is a statement about $\cos\alpha$, because cosine is decreasing on $[0, \pi]$. So it comes down to comparing two surds.

Solution

The two values of $\cos\alpha$

Write $c = \cos\alpha$. From the equation, $12\sin\alpha = 7 - 5c$. Squaring, and using $\sin^2\alpha = 1 - c^2$,

$$144(1-c^2) = 49 - 70c + 25c^2,$$
$$169c^2 - 70c - 95 = 0.$$

The quadratic formula gives

$$c = \frac{70 \pm \sqrt{70^2 + 4\cdot 169\cdot 95}}{2\cdot 169} = \frac{70 \pm \sqrt{69120}}{338}.$$

Since $69120 = 576 \times 120 = 24^2 \times 120$, we have $\sqrt{69120} = 24\sqrt{120}$. Hence

$$\cos\alpha = \frac{35 - 12\sqrt{120}}{169} \qquad \text{or} \qquad \cos\alpha = \frac{35 + 12\sqrt{120}}{169},$$

which is the required result, and the other value is $\dfrac{35 + 12\sqrt{120}}{169}$.

Ask yourself Squaring can create false solutions. Do both values really come from solutions of the original equation?

They do, and the single-cosine form shows why. Let β be the acute angle with $\cos\beta = \frac{5}{13}$ and $\sin\beta = \frac{12}{13}$. Then

$$5\cos x + 12\sin x = 13\cos(x-\beta),$$

so the equation says $\cos(x-\beta) = \frac{7}{13}$. Let γ be the acute angle with $\cos\gamma = \frac{7}{13}$, so that $\sin\gamma = \frac{\sqrt{169-49}}{13} = \frac{\sqrt{120}}{13}$. The solutions are $x = \beta \pm \gamma + 2k\pi$, and

$$\cos(\beta\pm\gamma) = \cos\beta\cos\gamma \mp \sin\beta\sin\gamma = \frac{35 \mp 12\sqrt{120}}{169}.$$

So each sign gives a genuine solution: $\alpha = \beta+\gamma$ gives the minus sign and $\alpha = \beta-\gamma$ gives the plus sign.

The second-quadrant solution

Suppose $\frac{1}{2}\pi < \alpha < \pi$. Then $\cos\alpha < 0$. Since $\sqrt{120} > 10$, we have $35 + 12\sqrt{120} > 0$, so this root is ruled out. Also $12\sqrt{120} > 120 > 35$, so the other root is negative. Therefore

$$\cos\alpha = \frac{35 - 12\sqrt{120}}{169}.$$

Cosine is strictly decreasing on $[0,\pi]$. Both α and $\frac{3}{4}\pi$ lie in this interval, so

$$\alpha < \tfrac{3}{4}\pi \iff \cos\alpha > \cos\tfrac{3}{4}\pi = -\tfrac{1}{\sqrt{2}}.$$

So we must show $\dfrac{12\sqrt{120}-35}{169} < \dfrac{1}{\sqrt{2}}$, that is, $12\sqrt{120} - 35 < \dfrac{169}{\sqrt{2}}$.

Ask yourself How do we compare these surds without a calculator, and without squaring something that might be negative?

We first bound $\sqrt{120}$ crudely. Because $120 < 121$, $\sqrt{120} < 11$, so

$$12\sqrt{120} - 35 < 132 - 35 = 97.$$

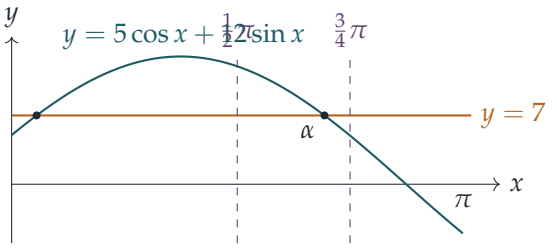
It is now enough to show $97 < \dfrac{169}{\sqrt{2}}$, that is, $97\sqrt{2} < 169$. Both sides are positive, so this is equivalent to the inequality between their squares:

$$2 \times 97^2 = 18\,818 < 28\,561 = 169^2,$$

which is true. Putting it together,

$$\cos\alpha = -\frac{12\sqrt{120}-35}{169} > -\frac{97}{169} > -\frac{1}{\sqrt{2}} = \cos\tfrac{3}{4}\pi,$$

and since cosine is decreasing on $[0,\pi]$, $\alpha < \frac{3}{4}\pi$, as required.



Checks

- Numerically, the roots in $[0, 2\pi)$ are $x \approx 0.1738$ and $x \approx 2.1782$. Their cosines are 0.98493 and -0.57073, which agree with $(35 \pm 12\sqrt{120})/169$.
- The second-quadrant root is 2.1782, which is less than $\frac{3}{4}\pi \approx 2.3562$, as the graph shows. The margin is comfortable: $\cos\alpha \approx -0.571$ against $-1/\sqrt{2} \approx -0.707$. That is why the crude bound $\sqrt{120} < 11$ is enough.
- As a sanity check on the quadratic, the two roots have product $-\frac{95}{169}$ and sum $\frac{70}{169}$.

Common slips

- Squaring an inequality without checking the signs. Here $12\sqrt{120} - 35$ and $169/\sqrt{2}$ are both positive, but the reason should be stated.
- Forgetting that cosine is *decreasing* on $[0, \pi]$, which reverses the inequality: $\alpha < \frac{3}{4}\pi$ corresponds to $\cos\alpha > -\frac{1}{\sqrt{2}}$.
- Not explaining why the positive root is excluded in the second quadrant.

Beyond STEP

The final part is a small exercise in rigorous estimation. You replace an awkward quantity by a simpler bound and check that the margin survives, which is how many inequalities in analysis are proved. The step from cosine values to angles uses the fact that a strictly monotonic function preserves (or reverses) order, which is the idea behind inverse functions in a first analysis course. For more on careful proof with inequalities, see Abbott, *Understanding Analysis*.

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